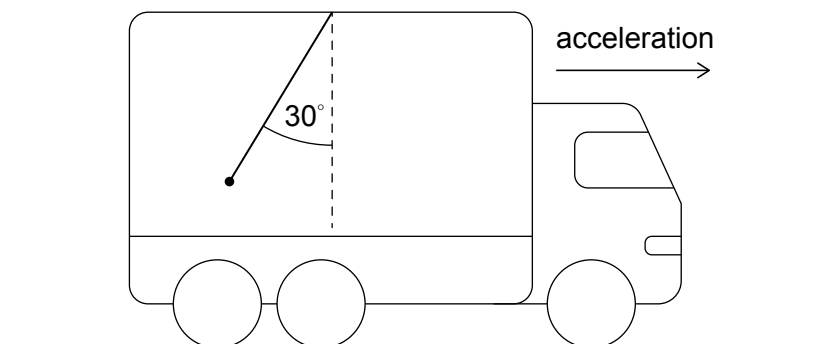


PAPER 2 / QUESTION 16

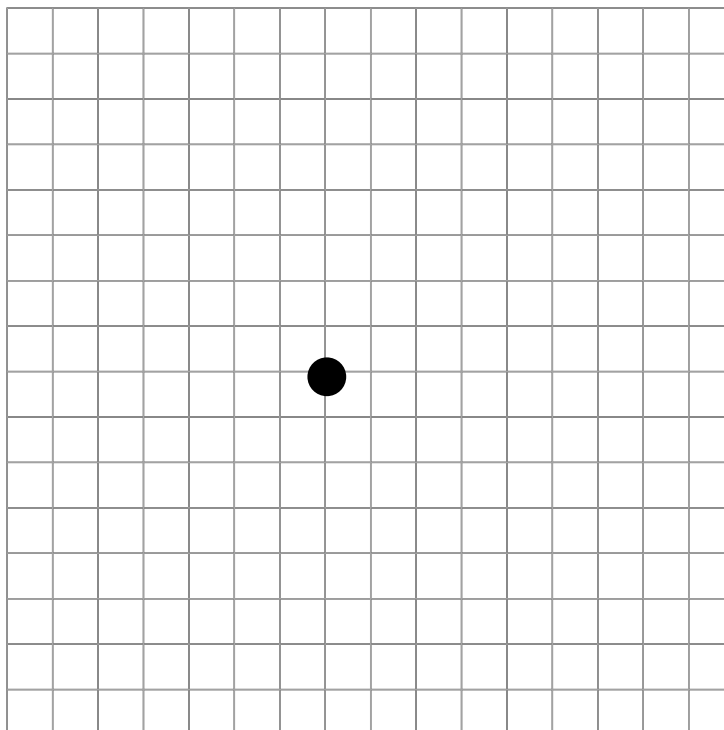
6 November 2025 / TZ3 (file label) / HL Paper 2 / original question 2 / 8825-6721

16. A ball is connected to a string that is attached to the ceiling of a truck. The truck accelerates to the right on a horizontal road. The string makes an angle of 30° with the vertical.



- (a) (i) Draw a labelled free-body diagram of the forces on the ball.

[2]



(This question continues on the following page)

PAPER 2 / QUESTION 16

6 November 2025 / TZ3 (file label) / HL Paper 2 / original question 2 / 8825-6721

(Question 16 continued)

(ii) Determine the acceleration of the truck.

[2]

.....

.....

.....

.....

(b) The truck now moves at **constant** speed up an incline that makes an angle of 10° to the horizontal. State and explain the angle between the string and the vertical.

[2]

.....

.....

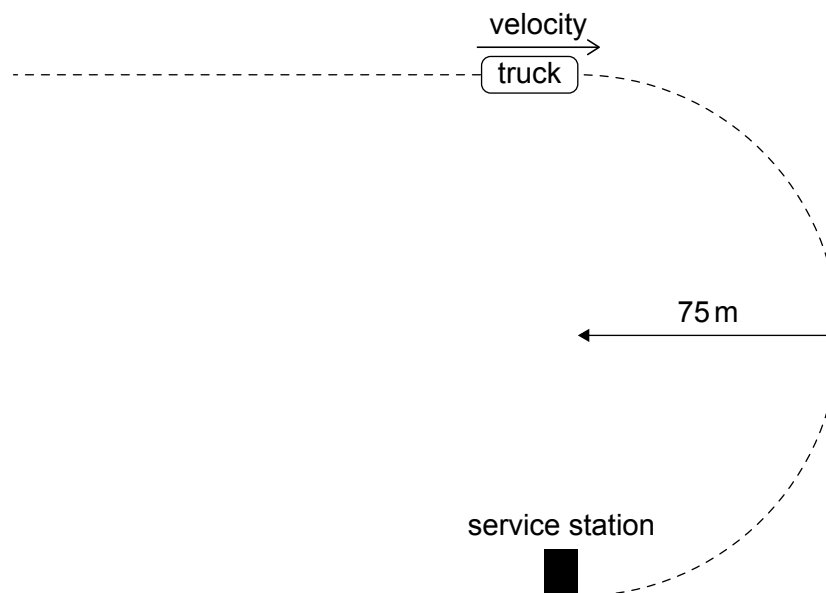
.....

.....

(This question continues on the following page)

(Question 16 continued)

On another occasion, the truck enters a horizontal semicircular road of radius of 75 m. The coefficient of static friction between the road and the tyres is 0.60.



- (c) Calculate the maximum speed at which the truck can travel on this road.

[2]

.....

.....

.....

.....

- (d) The truck runs out of fuel just as it enters the semicircle with speed 15 ms^{-1} . The resultant force opposing the motion has magnitude 520 N. The mass of the truck is 1400 kg. A service station is at the other end of the semicircle.

Determine whether the truck will be able to reach the service station.

[3]

.....

.....

.....

.....

.....

.....